homework03

P110-习题 8.4 (A 类) 3-7, 3-9; 9; 14-1, 14-3
P122-习题 8.6 (A 类) 7; 17;
P129-习题 8.7 (A 类) 1-1, 1-3; 3-2

3-7、隐函数求导 $x^3+y^3+z^3-3axyz=0$, 求 z_x, z_y, x_y

$$F_x=3x^2-3ayz \quad F_y=3y^2-3axz \quad F_z=3z^2-3axy$$

$$\Rightarrow z_x=\frac{-F_x}{F_z}=\frac{x^2-3ayz}{3axy-z^2}, \quad z_y=\frac{-F_y}{F_z}=\frac{y^2-3axz}{3axy-z^2}, \quad x_y=\frac{-F_y}{F_x}=\frac{y^2-3axz}{3ayz-x^2}$$

3-9、隐函数求导 $e^{x+y}\sin(x+z)=0$, 求 z_x, z_y, x_y

$$F_x=e^{x+y}\big(\sin(x+z)+\cos(x+z)\big), \quad F_y=e^{x+y}\sin(x+z), \quad F_z=e^{x+y}\cos(x+z)$$

$$\Rightarrow z_x=\frac{-F_x}{F_z}=-1-\tan(x+z), \quad z_y=\frac{-F_y}{F_z}=-\tan(x+z),$$

$$x_y=\frac{-F_y}{F_x}=\frac{-\sin(x+z)}{\sin(x+z)+\cos(x+z)}$$

9、设 $z=u^3+v^3$，而 u,v 由方程组 $\begin{cases} x=u+v, \\ y=u^2+v^2 \end{cases}$ 所确定，求 $\dfrac{\partial z}{\partial x}$，$\dfrac{\partial z}{\partial y}$。

$$\begin{cases} z_u = 3u^2 = z_x x_u + z_y y_u = z_x + 2u\, z_y, \\ z_v = 3v^2 = z_x x_v + z_y y_v = z_x + 2v\, z_y \end{cases}$$

$$\Rightarrow z_y = \frac{3(u^2-v^2)}{2(u-v)} = \frac{3}{2}(u+v) \Rightarrow z_x = \frac{3(u\,v^2 - v\,u^2)}{(u-v)} = -3\,uv$$

14-1、求方程组确定函数的导数 $\begin{cases} z=x^2+y^2 \\ x^2+2y^2+3z^2=20 \end{cases}$，求 $y'(x), z'(x)$

$$\begin{cases} z' = 2x + 2y\,y' \\ 2x + 4y\,y' + 6z\,z' = 0 \end{cases} \Rightarrow \begin{cases} y' = \frac{-(2x+6z\,2x)}{4y+6z\,2y} = \frac{-x(1+6z)}{y(2+6z)} \\ z' = 2x + 2y\frac{-x(1+6z)}{y(2+6z)} = \frac{x}{1+3z} \end{cases}$$

14-3、求方程组所确定的函数 $\begin{cases} x = e^u + u\sin v, \\ y = e^u - u\cos v, \end{cases}$ 求 $\dfrac{\partial u}{\partial x}$, $\dfrac{\partial u}{\partial y}$, $\dfrac{\partial v}{\partial x}$, $\dfrac{\partial y}{\partial v}$。

$$\begin{cases} dx = \left(e^u + \sin v\right) du + u\,\cos v\, dv \\ dy = \left(e^u - \cos v\right) du + u\,\sin v\, dv \end{cases}$$

$$\left((e^u + \sin v)\sin v - (e^u - \cos v)\cos v\right) du = \sin v\, dx - \cos v\, dy,$$

$$u\left(\cos v(e^u - \cos v) - \sin v(e^u + \sin v)\right) dv = (e^u - \cos v)\, dx - (e^u + \sin v)\, dy$$

$$\Rightarrow u_x = \frac{\sin v}{e^u(\sin v - \cos v) + 1}, \quad u_y = \frac{-\cos v}{e^u(\sin v - \cos v) + 1},$$

$$v_x = \frac{e^u - \cos v}{u\left(e^u(\cos v - \sin v) - 1\right)}, \quad v_y = \frac{-e^u - \sin v}{u\left(e^u(\cos v - \sin v) - 1\right)}$$

7、求 $f(x,y,z)=x^2+y^2+z^2$ 在椭球面 $\frac{x^2}{a^2}+\frac{y^2}{b^2}+\frac{z^2}{c^2}=1$ 上的点 $P(x_0,y_0,z_0)$ 的外法线方向的导数.

解: 依题意,

$$\nabla f=(2x,2y,2z),\quad n_{法}=(\frac{2x}{a^2},\frac{2y}{b^2},\frac{2z}{c^2})$$

$$\Rightarrow f_l=(\frac{4x^2}{a^2}+\frac{4y^2}{b^2}+\frac{4z^2}{c^2})\Big/\sqrt{\frac{4x^2}{a^4}+\frac{4y^2}{b^4}+\frac{4z^2}{c^4}}=2\Big/\sqrt{\frac{x^2}{a^4}+\frac{y^2}{b^4}+\frac{z^2}{c^4}}$$

17、求函数 $z=\ln\frac{y}{x}$ 分别在点 $A(\frac{1}{3},\frac{1}{10})$ 和 $B(1,\frac{1}{6})$ 处的两个梯度的夹角余弦.

解: 依题意 $\Delta z=(\frac{-1}{x},\frac{1}{y})$

$$\Rightarrow v_1=\big(-3,10\big),\quad v_2=\big(-1,6\big)$$

$$\Rightarrow\cos\theta=\frac{v_1\cdot v_2}{|v_1||v_2|}=\frac{63}{\sqrt{109}\sqrt{37}}=\frac{32}{\sqrt{4033}}$$

1-1、求指定的切线方程和法平面方程

$x = t - \cos t, y = 3 + \sin 2t, z = 1 + \cos 3t$, 点 $t = \frac{\pi}{2}$

解: 依题意 $t = \frac{\pi}{2}$ 时, $x = \frac{\pi}{2}, y = 3, z = 1$

$$v = \left(x'(t), y'(t), z'(t)\right)\Big|_{t=\frac{\pi}{2}} = \left(1 + \sin t, 3 - 2\cos 2t, -3\sin 3t\right)\Big|_{t=\frac{\pi}{2}} = (2, 5, 3)$$

$$\Rightarrow l_{\text{切}} : \frac{x - \pi/2}{2} = \frac{y - 3}{5} = \frac{z - 1}{3}, \quad \pi_{\text{法}} : 2x + 5y + 3z = \pi + 18$$

1-3、求指定的切线方程和法平面方程 $x^2 + z^2 = 10, y^2 + z^2 = 10$, 点 $(1, 1, 3)$

解: 依题意

$$n_1 = (2x, 0, 2z)\big|_{(1,1,3)} = (2, 0, 6), \quad n_2 = (0, 2y, 2z)\big|_{(1,1,3)} = (0, 2, 6)$$

$$\Rightarrow v = n_1 \times n_2 = (-12, -12, 4) \Rightarrow l_{\text{切}} : \frac{x - 1}{3} = \frac{y - 1}{3} = \frac{z - 3}{-1}$$

$$\Rightarrow \pi_{\text{法}} : 3x + 3y - z = 3$$

3-2、求下列曲面在指定点的切平面方程与法线方程 $x^2+y^2+z^2=169$, 点 $(3,4,12)$;

解: 依题意

$$n_{法}=(2x,2y,2z)\big|_{(3,4,12)}=(6,8,24)\Rightarrow \pi_{切}:3x+4y+12z=169,$$

$$l_{法}:\frac{x-3}{3}=\frac{y-4}{4}=\frac{z-12}{12}$$